



LoRa Alliance
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LoRaWAN End-Device Specification for SCHC Certification

Version 1.1

Table of Contents

1. Introduction	7
2. Test Procedure	8
2.1 Compression rules	8
• RuleIDx: no-compression	8
• RuleIDy: compression	8
• RuleIDz: compression.....	9
2.2 Fragmentation RuleID	9
2.3 Header configuration	10
2.4 Pre-routes	11
3. Test Cases.....	11
3.1 Pre-Requisite Setup Test Cases	11
3.1.1 Test Procedure Frame Sequence Chart	12
3.2 SCHC message, RuleID and IID computation	13
3.2.1 Test Procedure Frame Sequence Chart	14
3.3 Compression mechanisms	14
3.3.1 Test Procedure Frame Sequence Chart	15
3.4 Decompression mechanisms	16
3.4.1 Test Procedure Frame Sequence Chart	16
3.5 Uplink Fragmentation (fixed MTU)	17
3.5.1 Test Procedure Frame Sequence Chart	19
3.6 Uplink Fragmentation (MTU changing)	20
3.6.1 Test Procedure Frame Sequence Chart	20
3.7 Uplink Fragmentation-Retransmission (fixed MTU)	21
3.7.1 Test Procedure Frame Sequence Chart	22
3.8 Uplink Fragmentation-Retransmission (variable MTU)	23
3.8.1 Test Procedure Frame Sequence Chart	24
3.9 Uplink Fragmentation- ACK Request retransmission	25
3.9.1 Test Procedure Frame Sequence Chart	26
3.10 Uplink Fragmentation-Sender Abort (no SCHC ACK)	27
3.10.1 Test Procedure Frame Sequence Chart	27

3.11	Uplink Fragmentation-Sender Abort (all tiles received but incorrect RCS)	28
3.11.1	Test Procedure Frame Sequence Chart	29
3.12	Uplink Fragmentation-Receiver Abort	30
3.12.1	Test Procedure Frame Sequence Chart	30
3.13	Uplink-Compression and fragmentation	31
3.13.1	Test Procedure Frame Sequence Chart	32
3.14	Downlink fragmentation- reassembly	33
3.14.1	Test Procedure Frame Sequence Chart	33
3.15	Downlink fragmentation-Receiver abort	34
3.15.1	Test Procedure Frame Sequence Chart	35
3.16	Downlink fragmentation-Lost Ack	35
3.16.1	Test Procedure Frame Sequence Chart	36
3.17	Downlink fragmentation (reassembly) and decompression	38
3.17.1	Test Procedure Frame Sequence Chart	38

Revision History

Version	Author	Updates	Date
1.0 Draft A	Hussein Al Haj Hassan- Nisha Bhaskaran Laurent Toutain- Ivan Martinez- Miguel Luis Julien Lecoeuvre- Remi Demerle- Diego Bartolome- Michael Kirwan- Jerome Elias	Initial Version 1.0	May 12, 2022
1.0 Draft B	Hussein Al Haj Hassan	-Editorial updates -Reference of TS010 is added -Pre-requisite setup test cases added -All Tests procedures and description are updated to follow the proposed updated of TS09 and other LoRa Alliance specification documents -Downlink illegal RuleID test is removed -Some Typos are corrected	July 28 2022
1.0	Nisha Bhaskaran	Approved by CC	September 21 2022
1.1	Hussein Al Haj Hassan Nisha Bhaskaran	-Clarifying how a compression RuleID is selected -Clarifying the behavior not yet included in the test plan -Clarifying that All-1 message is counted in MAX_ACK_REQ -Clarifying the accepted behaviors in test 3.1t5	December 21 2022



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Glossary

LoRaWAN	Protocol specification developed and maintained by the LoRa Alliance.
Certification Logo	LoRa Alliance defined logo that can be displayed on the Certified product and any documentation and marketing information about the End-Device.
LoRa Test House	Organization and corresponding facility accredited by the LoRa Alliance to perform Certification testing.
End Device	Device submitted to a LoRa-Alliance Authorized Test House (ATH) for Certification.

Abbreviations & Acronyms

ABP	Activation by Personalization
ADR	Adaptive Data Rate
AS	Application Server
ATH	Authorized Test House
AWG	Arbitrary Waveform Generator
DR	Data Rate
DUT	Device Under Test
ERP	Equivalent Radiated Power compared to a dipole antenna (expressed in dBd)
EIRP	Equivalent Isotropic Radiated Power: $ERP = EIRP - 2.15\text{dB}$ (expressed in dBi)
ETSI	European Telecommunications Standards Institute
FSK	Frequency Shift Keying modulation technique.
GW	Gateway
LCTT	LoRaWAN Conformance Test Tool



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MAC	Media Access Control
NS	Network Server
OTAA	Over-the-Air Activation
TCL	Test Control Layer
TRP	Total Radiated Power

Definitions of terms used in this document

XXXX	X is a valid number which would vary based on the region being tested
DC	Dynamic Channel
FC	Fixed Channel
MinDR	Minimum Data Rate for the region tested, as specified in the Regional Parameters Specification RP2 1.0.1
Max125kHzDR	Maximum Data Rate using 125 kHz bandwidth for the region tested, as specified in the Regional Parameters Specification RP2 1.0.1
R	Repeat until the condition specified within brackets [] is completed

References

[1]	LoRaWAN Specification TS001 L2 v1.0.4.
[2]	LoRaWAN Regional Parameters Specification RP002 v1.0.1.
[3]	LoRaWAN Certification Protocol Specification TS009 v1.1.0.
[4]	LoRaWAN End Device Certification Requirements for All Regions v1.5
[5]	LoRaWAN® IPv6 Adaptation Layer Specification TS010-1.0.0
[6]	[RFC791] “Internet Protocol- DARPA internet program protocol specification”, RFC 791, 1981

1. Introduction

This document specifies the minimum testing requirements for an End-Device to be designated as “**SCHC over LoRaWAN Certified**”. SCHC over LoRaWAN certification will confirm that the End-Device meets the functional requirements of the LoRaWAN IPv6 Adaptation Layer Specification TS010-1.0.0 [5]. Note that the End-Device must already be “LoRaWAN certified” as a prerequisite for the SCHC over LoRaWAN certification as defined in the LoRaWAN End Device Certification Requirements for All Regions v1.5 [4].

The Functional Test Description for SCHC over LoRaWAN Certification is the same used for LoRaWAN certification presented in [3].

The proposed DUT under test architecture is as follows:

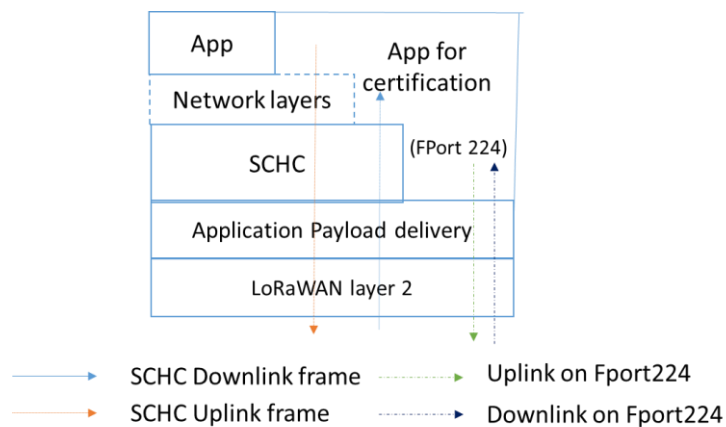


Figure 1 DUT Architecture

The architecture does not necessarily show all the layers of the DUT, but rather focuses on the implementation related to certification of SCHC, i.e., the IPv6 adaptation layer.

“App” represents the device application that uses SCHC in real operation.

“App for certification” represents the application that SHALL be implemented on the DUT to be able to test the SCHC implementation. FPort 224 will be used for receiving the commands from the TCL and calling appropriate functions in the SCHC implementation. FPort 224 is also used as feedback for some tests as explained later. This application SHALL be disabled on any device in production, otherwise it may be intentionally or accidentally be activated to harm the device itself or the networks in its radio coverage. The DUT SHALL be sent to the Authorized Test House (ATH) with the FPort enabled for the Certification test, then returned to the End-Device manufacturer for the FPort to be disabled

“Network layers” or third party libraries can be used to generate the IPv6 and UDP headers. They may be implemented within the app for certification. This document does not specify how these layers are implemented.

The tests are based on a sequence of exchanges with the TCL. The TCL provides the DUT with inputs and commands. The DUT SHALL process the input and reply correspondingly.

2. Test Procedure

The certification will take place as set of interactions between the TCL and the DUT. Commands are sent from the TCL on a specific port (FPort 224). Each command will contain all the necessary inputs to execute the command. When the DUT receives a message with FPort 224, the application payload delivery will forward it to “App for Certification”.

Nevertheless, some SCHC messages will be sent by the TCL to the DUT on specific FPorts and will be part of the certification process as explained later in this document.

Note that the fields of SCHCACKInd message and the SCHC messages on FPort other than 224 do not have octet definition. The fields of these messages will be represented as network order as specified in RFC791 [6].

The RuleIDs that are involved with SCHC certification are: RuleIDx, RuleIDy, RuleIDz, FPortup, and FPortDown. They are explained below.

2.1 Compression rules

The tester SHALL implement the following compression/ decompression rules

- **RuleIDx: no-compression**

This RuleID corresponds to the case where none of the headers (UDP/IPv6) are compressed. The recommended (default) value of FPort is 22.

- **RuleIDy: compression**

The default value of RuleIDy is 100.

This rule compresses/ decompresses the headers (of UDP/IPv6) as follows:

FID	FL	F P	DI	TV	MO	CDA	Sent (bits)
IPv6 version	4	1	Bi	6	Equal	Not-sent	
IPv6 Diffserv	8	1	Bi	0	Ignore	Not-sent	
IPv6 Flow label	20	1	Bi	0	ignore	Not-sent	
IPv6 length	16	1	Bi		Ignore	Compute-*	
IPv6 next header	8	1	Bi	17	Equal	Not-sent	
IPv6 Hop limit	8	1	Up	40	Ignore	Not-sent	
IPv6 DevPrefix	64	1	Bi	Fe80::/64 Aaaa::/64 Bbbb::/64	Match- mapping	Mapping sent	2
IPv6 DevIID	64	1	Bi		Ignore	DevIID	
IPv6 AppPrefix	64	1	Bi	Aaaa::/64 Bbbb::/64	Match- mapping	Mapping sent	2

				cccc::/64 Fe80::/64			
IPv6 AppIID	64	1	Bi	::1	Equal	Not sent	
UDP DevPort	16	1	Bi	0x5C40	MSB(12)	LSB (4)	4
UDP AppPort	16	1	Bi	0x1070	MSB (12)	LSB (4)	4
UDP length	16	1	Bi		Ignore	Compute-*	
UDP checksum	16	1	Bi		Ignore	Compute-*	

FID: Field identifier; FL: Field length (in bits) FP: Field position; DI: Direction; TV: Target value; MO: Matching operator; CDA: Compression/decompression action

- **RuleIDz: compression**

The default value of RuleIDz is 101.

This rule compresses/ decompresses the headers (of UDP/IP) as follows:

FID	FL	FP	DI	TV	MO	CDA	Sent (bits)
IPv6 version	4	1	Bi	6	Equal	Not-sent	
IPv6 Diffserv	8	1	Bi	0	Ignore	Not-sent	
IPv6 Flow label	20	1	Bi	0	ignore	Not-sent	
IPv6 length	16	1	Bi		Ignore	Compute-*	
IPv6 next header	8	1	Bi	17	Equal	Not-sent	
IPv6 Hop limit	8	1	Dwn	1	Ignore	Not-sent	
IPv6 DevPrefix	64	1	Bi	Aaaa::/64 Fe80::/64	Match- mapping	Mapping sent	1
IPv6 DevIID	64	1	Bi		Ignore	DevIID	
IPv6 AppPrefix	64	1	Bi	Aaaa::/64 Bbbb::/64 Eeee::/64 Fe80::/64	Match- mapping	Mapping sent	2
IPv6 AppIID	64	1	Bi	::1	Equal	Not sent	
UDP DevPort	16	1	Bi	0x5C40	MSB(12)	LSB (4)	4
UDP AppPort	16	1	Bi	0x2070	MSB (12)	LSB (4)	4
UDP length	16	1	Bi		Ignore	Compute-*	
UDP checksum	16	1	Bi		Ignore	Compute-*	

For RuleIDx, RuleIDy and RuleIDz, in case the DUT has the compression/ decompression rule with a RuleID that is different than the default mentioned above, the tester SHALL change the value of RuleID when configuring the TCL to the one used in his/her implementation.

Note: The tester MAY have a SCHC implementation that can re-configure the compression rules. Configuring the compression rules SHALL be done before testing and the method of configuring is out of scope of the certification process.

2.2 Fragmentation RuleID

The following fragmentation rules SHALL be implemented

a) FPortUp

This RuleID corresponds to the Uplink Fragmentation, which uses Ack-on-Error mechanism. The default (recommended) RuleID for uplink fragmentation is 20.

b) FPortDown

This RuleID corresponds to the Unicast Downlink Fragmentation, which uses Ack-Always mechanism. The default (recommended) RuleID for Downlink fragmentation is 21.

For FPortUp and FPortDown, in case the tester implemented this Fragmentation with a RuleID that is different than the default one mentioned above, the tester SHALL change the value of FPortUp when configuring the TCL to the one used in his/her implementation.

2.3 Header configuration

Unless explicitly mentioned, the DUT SHALL use the following IPv6/UDP headers and send packets with headers constructed correspondingly:

- UDP Dev Port: 0x5C40 (2 bytes)
- UDP App Port: 0x3070 (2 bytes)
- UDP length field: computed (2 bytes)
- UDP checksum: computed (2 bytes)
- IPv6 Version field: 6 (4 bits)
- IPv6 Traffic class: 0 (8 bits)
- IPv6 Flow label: 0 (20 bits)
- IPv6 Payload length: computed (2 bytes)
- IPv6 Next header field: 17 (8 bits)
- IPv6 Hop limit field: 255 (8 bits)
- IPv6 device prefix: fe80::/64 (64 bits)
- IPv6 device IID (64 bits): computed based on provided DevEUI and AppSKey
- IPv6 App prefix: fe80::/64 (64 bits)
- IPv6 App IID: ::1 (64 bits)

Extension headers SHALL not be used.

During certification, all these fields must be tested (when applicable), except IPv6 Traffic class, IPv6 Flow label and IPv6 Hop limit field.

During uplink tests, the DUT determines which compression RuleID to use by applying the matching operators to UDP/IPv6 fields. In particular, the UDP App Port may vary depending on the messages received from the LCTT. During downlink tests, the decompression RuleID is the one sent by the TCL.

2.4 Pre-routes

During certification, any frame received by the DUT with FPort=224 SHALL be forwarded to the App for certification.

Other downlink messages received by DUT will be transferred to the App for certification after being processed by the SCHC implementation.

3. Test Cases

3.1 Pre-Requisite Setup Test Cases

The Pre-requisite setup tests need to be run as a first step on the DUT before performing any of the other test cases mentioned in this document, to speed up the test execution.

The DUT is brought to the appropriate state to conduct the Relay tests by executing these test cases.

The Pre-requisite tests include-

1. Change TXPeriodicity to 5 seconds
2. Switch to Unconfirmed uplink frames (if not already)
3. Turn on the ADR bit (if not already set)
4. Change the uplink DataRate to Max125KHz

3.1.1 Test Procedure Frame Sequence Chart

Step	Procedure	Frame Sequence		Test Purpose
		End Device - TCL	Frame	
1	DUT sends Confirmed or Unconfirmed frame	→		
	The TCL sends Unconfirmed frame in RX1 window	←	CP-CMD TxPeriodicityChangeReq FPort = 224 Periodicity = 5 Payload = [0x]0601 If uplink was a Confirmed frame, send ACK	Uplink Periodicity set to 5s
2	DUT sends Confirmed or Unconfirmed frame	→		
	If DUT sent a Confirmed frame, then The TCL sends Unconfirmed frame Else, this step must be skipped	←	CP-CMD TxFramesCtrlReq FPort = 224 Frame type = Unconfirmed Payload = [0x]0701 If DUT sent Confirmed uplink frame, TCL must Acknowledge	Switch to Unconfirmed frames
3	DUT sends Unconfirmed frame	→		
	If FCtrl ADR Bit = false, then The TCL sends Unconfirmed frame Else, this step is skipped	←	CP-CMD ADRBitChangeReq-ON FPort = 224 Payload = [0x]040	
4	DUT sends Unconfirmed frame	→	FCtrl ADR bit = true	Turn on the ADR bit
	The TCL sends Unconfirmed frame	←	MAC-CMD LinkADRRReq DataRate = Max125kHzDR, refer [2] Payload = [0x]03XXXXXXXXXX ChMaskCntl: DC = 0, FC = 6 ChMask: DC - Enable only default channels FC = [0x]00FF	For FC device using 8-channel gateway, refer to Section 2 of [4] for ChMask settings
5	DUT sends Unconfirmed frame	→	MAC-CMD LinkADRAns Payload = [0x]0307	Changed the DR to Max125kHz DR

3.2 SCHC message, RuleID and IID computation

Test Purpose:

Test the capability of the DUT to compose a SCHC message, send non-compressed message and calculate the IID.

Test Description:

Input: The TCL sends the DUT a CP-CMD SCHCMsgSendReq to ask the DUT to send an uplink message with a specific data size (1 bytes) and 0x3070 as a UDP App Port.

Expected Output:

The DUT must reply with the SCHCMsgSendAns.

Then, the DUT must then send a non-compressed packet with FPort=RuleIDx (default is 22) to the TCL.

Fport	LoRaWAN payload		
RuleIDx (1 byte)	IPv6 header (40 bytes)	UDP header (8 bytes)	Data (as per command)

Figure 2 Expected reply for the SCHCMessageSend command with 0x3070 as UDP App Port

The SCHC message from the DUT must be composed of the RuleID, placed in the FPort, and the LoRaWAN payload. The RuleID (FPort) for this test is RuleIDx corresponding to the no-compression. The LoRaWAN payload consists of the IPv6/UDP headers (no compression is applied) and the data to be sent (bits are zeros).

Verify

- 1) RuleID (FPort)=RuleIDx
- 2) Correct UDP/IPv6 headers, where DevIID is correctly computed based on the Application session key (AppSkey) and the DevEUI as specified by RFC9011
- 3) Size of the LoRaWAN payload (when the data is 1 bytes the LoRaWAN payload is 48+1=49 bytes)

3.2.1 Test Procedure Frame Sequence Chart

Step	Procedure	Frame Sequence		Test Purpose
		End Device - TCL	Frame	
1	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	MAC-CMD LinkADRReq DataRate = Max125kHzDR, refer [2] , Payload = [0x]03XXXXXXXX	
2	DUT sends Unconfirmed frame	→	MAC-CMD LinkADRAns Payload = [0x]0307	
	TCL sends Unconfirmed frame	←	CP-CMD SCHCMsgSendReq FPort = 224 Payload = [0x]5070300100	Test the ability to compose a SCHC message and calculate the DevIID
3	DUT sends Unconfirmed frame	→	CP-CMD SCHCMsgSendAns FPort = 224 Payload = [0x]50	
4	DUT sends Unconfirmed frame	→	No-comp SCHC packet contains - FPort = RuleIdx - Payload = IPv6/UDP headers (with computed DevIID, checksum) & additional data of specified size	
5	DUT sends Unconfirmed frame	→		

3.3 Compression mechanisms

Test Purpose:

Test the capability of the DUT to apply header compression

Test Description:

Input:

The TCL sends the DUT a CP-CMD SCHCMsgSendReq to ask the DUT to send an uplink message with a specific data size (1 byte) and 0x1070 as a UDP App Port. The payload of this message will be the input for the DUT.

Expected Output:

The DUT must reply with the SCHCMsgSendAns.

Then, The DUT must then send a compressed packet with

- FPort=RuleIDy (default is 100)
- Compression residue
- data of specified size in the SCHCMsgSendReq
- Padding (all zeros)

Fport		LoRaWAN payload	
RuleIDy (1 byte)	Compression residue	Data (as per command)	Padding

Figure 3 Expected reply for the SCHCMessageSend command with 0x1070 as UDP App Port

3.3.1 Test Procedure Frame Sequence Chart

Step	Procedure	Frame Sequence		Test Purpose
		End Device - TCL	Frame	
1	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	MAC-CMD LinkADRReq DataRate = Max125kHzDR, refer [2], Payload = [0x]03XXXXXXXXXX	
2	DUT sends Unconfirmed frame	→	MAC-CMD LinkADRAns Payload = [0x]0307	
	TCL sends Unconfirmed frame	←	CP-CMD SCHCMsgSendReq FPort = 224 Payload = [0x]5070100100	Test Compression
3	DUT sends Unconfirmed frame	→	CP-CMD SCHCMsgSendAns FPort = 224 Payload = [0x]50	

4	DUT sends Unconfirmed frame	→	Compressed SCHC packet contains - FPort = RuleIDy (default 100) - Payload = compression residue (computed), additional payload of specified size and padding	
5	DUT sends Unconfirmed frame	→		

3.4 Decompression mechanisms

Test Purpose:

Test the capability of the DUT to apply decompression.

Test Description

Input: The TCL sends a compressed SCHC packet to the DUT with FPort = RuleIDz, Compression residue, Data (1 bytes, with zero bits) and Padding (5 bits of zero).

The Compression Residue = b'11100000000

Expected Output: A decompressed-header packet using RuleIDz must be sent back to the TCL via FPort 224 (FPort 224 is used as feedback) with same Data 1 bytes using the SCHCACKInd message.

3.4.1 Test Procedure Frame Sequence Chart

Step	Procedure	Frame Sequence		Test Purpose
		End Device – TCL	Frame	
1	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	MAC-CMD LinkADRReq DataRate = Max125kHzDR, refer [2] Payload = [0x]03XXXXXXXXX	

2	DUT sends Unconfirmed frame	→	MAC-CMD LinkADRAAns Payload = [0x]0307	
	TCL sends Unconfirmed frame	←	Compressed SCHC packet contains - FPort = RuleIDz - Payload = Compression residue = b'11100000000 Data = 1 byte of zero Padding = 5 bits of zeros	Test the capability of a device to decompress data
3	DUT sends Unconfirmed frame	→	CP-CMD SCHCACKInd FPort = 224 - Payload = [0x]51 +IPv6 and UDP headers and Data of 1 byte	
4	DUT sends Unconfirmed frame	→		

3.5 Uplink Fragmentation (fixed MTU)

Test Purpose:

Test the capability of the DUT to apply uplink Fragmentation (Ack-on-error) with fixed MTU.

Test Description

Input: The TCL sends the DUT a CP-CMD SCHCMsgSendReq to ask the DUT to send an uplink message with a large data size = 650 bytes and 0x3070 as a UDP App Port.

Expected Output:

The DUT must reply with the SCHCMsgSendAns.

Then,

A non-compressed SCHC packet with FPort=RuleIDx (default is 22) must be sent back to the TCL via several fragments. The DUT must apply fragmentation and send the fragments following Ack-on-error mechanism.

The number and size of fragments depend on the region and DR.

Note: The chosen Data size will enforce fragmentation even at the highest DR.

Fragmentation

The SCHC message contains:

- RuleID: RuleIDx (i.e. no compression)

- UDP and IP headers
- Data of number of bytes specified in the command (650). The data bits are zeros.

For all fragments: FPort = FPortUp (default of 20)

The LoRaWAN payload in the fragment depends on the MTU (region)

First fragment (order and content):

- W field on 2 bits: value is 0
- FCN field on 6 bits: value is 62
- Remaining payload: size is multiple of 10

(First tile: no-compression RuleIDx+9 byte. The number of other tiles depend on the MTU)

Intermediate Fragments:

- W (value depends on the window): The value is incremented by 1 after having 63 tiles (630 bytes)
- FCN (value depends on the first tile of the fragment)
- Tiles (number of tiles related to MTU)

Last Fragment (and the one before):

- The last tile can be sent in a regular SCHC fragment or in All-1 message. The mechanism used will be specified in the Certification Questionnaire.
- W (value depends on the window): The value is incremented by 1 after having 63 tiles (630 bytes)
- FCN is ones in the all-1 fragment
- RCS should be checked (after receiving all the tiles)

Verify

- After receiving the Last fragment, the TCL must check if the size of the SCHC message after fragmentation is correct.
- In case the size of data is 650 bytes, the SCHC message must be 1 byte (RuleIDx)+48 bytes (IPv6+UDP header)+650 bytes of data.

Note: There are two possibilities to send SCHC Acks during uplink fragmentation- either at the end of each window or until the end of all windows (after All-1 message is sent). The current tests only accept the case where the bitmap is sent after the All-1 message. This is applied for all the tests with uplink fragmentation.

3.5.1 Test Procedure Frame Sequence Chart

Step	Procedure	Frame Sequence		Test Purpose
		End Device - TCL	Frame	
1	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	MAC-CMD LinkADRRReq DataRate = Max125kHzDR, refer [2] Payload = [0x]03XXXXXXXXXX	
2	DUT sends Unconfirmed frame	→	MAC-CMD LinkADRAns Payload = [0x]0307	
	TCL sends Unconfirmed frame	←	CP-CMD SCHCMsgSendReq FPort = 224 Payload = [0x]5070308A02	Test DUT's Fragmentation mechanism
3	DUT sends Unconfirmed frame	→	CP-CMD SCHCMsgSendAns FPort = 224 Payload = [0x]50	
4	DUT sends Unconfirmed frame	→	First fragment contains: - FPort = FPortUp - Payload = W (2 bits) = 0 FCN (6 bits) = 62 RuleIdx Remaining bytes from IPv6 header, UDP header and data	
5	DUT sends Unconfirmed frames	→ R [until last fragment]	Fragmentation session continues, next (not last) fragment: - FPort = FPortUp - Payload = W=computed, FCN=computed, and Remaining bytes from IPv6 header, UDP header and data This fragment type is repeated until the last fragment	
6	DUT sends Unconfirmed frame	→	All-1 Fragment (Last fragment), - FPort = FPortUp - Payload = W=computed, FCN=b'111111, RCS=computed Note: last tile can be held on regular SCHC fragment or in the All-1	

	TCL sends Unconfirmed frame	←	SCHC Ack RuleID: FPortUp W= value of last window C=1 Padding	
7	DUT sends Unconfirmed frame	→		

3.6 Uplink Fragmentation (MTU changing)

Test Purpose:

Test the capability of the DUT to apply uplink Fragmentation (Ack-on-error) with variable MTU.

Test Description

This test is the same as Uplink fragmentation (fixed MTU) Section 3.5, with the difference of changing the MTU during the fragmentation process. Here, the FOpts will appear as result of the MAC command.

3.6.1 Test Procedure Frame Sequence Chart

Step	Procedure	Frame Sequence		Test Purpose
		End Device - TCL	Frame	
1	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	MAC-CMD LinkADRRReq DataRate = Max125kHzDR, refer [2] Payload = [0x]03XXXXXXXX	
2	DUT sends Unconfirmed frame	→	MAC-CMD LinkADRAns Payload = [0x]0307	
	TCL sends Unconfirmed frame	←	CP-CMD SCHCMsgSendReq FPort = 224 Payload = [0x]5070308A02	
3	DUT sends Unconfirmed frame	→	CP-CMD SCHCMsgSendAns FPort = 224 Payload = [0x]50	
4	DUT sends Unconfirmed frame	→	First fragment: FPortUp in FPort, W=0, FCN=62, RuleIDx, bytes from (IPv6 header, UDP header, Data)*	

	The TCL sends Unconfirmed frame	←	MAC-CMD LinkADRReq DataRate = Max125kHzDR-2, refer [2] Payload = [0x]03XXXXXXXX	
5	DUT sends Unconfirmed frames	→ R [until last fragment]	MAC-CMD LinkADRAns (in FOpts field) Payload = [0x]0307 Next (not last) fragment: FPortUp in Fport, W=computed, FCN=computed, and bytes from (IPv6 header, UDP header, Data)* This fragment type is repeated until the last fragment Note: The MAC-CMD LinkADRAns could also be split and sent as a separate uplink first, followed by the SCHC fragment in the next uplink. Both methods are acceptable.	
	The TCL sends Unconfirmed frame	←	MAC-CMD LinkADRReq If DR = Max125kHzDR, change to Max125kHzDR-2 If DR = Max125kHzDR-2, change to Max125kHzDR Payload = [0x]03XXXXXXXX	
6	DUT sends Unconfirmed frame	→	All-1 Fragment (last fragment) FPortUp in Fport, W=computed, FCN=b'111111, RCS=computed The last tile can be held on regular SCHC fragment or in the All-1	
	TCL sends Unconfirmed frame	←	SCHC ACK RuleID: FPortUp W= value of last window C=1 Padding	
7	DUT sends Unconfirmed frame	→		

3.7 Uplink Fragmentation-Retransmission (fixed MTU)

Test Purpose:

Test the capability of the DUT to retransmit when there is error/missing when the MTU is not changing.

Test Description

This test is the same as Uplink fragmentation (fixed MTU) Section 3.5, but it takes into consideration the possibility of losing some tiles. The TCL will reply to the DUT by sending a bitmap.

Note: There are two possibilities for SCHC to send a bitmap- either at the end of each window or until the end of all windows. This is determined by the implementer. Here, the one that sends at the end is used, as it works with both.

3.7.1 Test Procedure Frame Sequence Chart

Step	Procedure	Frame Sequence		Test Purpose
		End Device - TCL	Frame	
1	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	MAC-CMD LinkADRReq DataRate = Max125kHzDR, refer [2] Payload = [0x]03XXXXXXXXX	
2	DUT sends Unconfirmed frame	→	MAC-CMD LinkADRAns Payload = [0x]0307	
	TCL sends Unconfirmed frame	←	CP-CMD SCHCMsgSendReq FPort = 224 Payload = [0x]5070308A02	Test DUT's Fragmentati on mechanism
3	DUT sends Unconfirmed frame	→	CP-CMD SCHCMsgSendAns FPort = 224 Payload = [0x]50	
4	DUT starts Unconfirmed frame	→	First fragment: FPortUp in FPort, W=0, FCN=62, RuleIdx, bytes from (IPv6 header, UDP header, Data)*	
5	DUT sends Unconfirmed frame	→ R [until the last fragment]	Next (not last) fragment: FPortUp in Fport, W=computed, FCN=computed, and bytes from (IPv6 header, UDP header, Data)	
6	DUT sends Unconfirmed frame	→	All-1 Fragment (last fragment) FPortUp in Fport, W=computed, FCN=b'111111, RCS=computed The last tile can be held on regular SCHC fragment or in the All-1	

	TCL sends Unconfirmed frame	←	SCHC ACK-incorrect RCS -RuleID=FPortUp -W(2 bits)=0 -C (1 bit)=0 -Compressed bitmap: binary depends on the tiles missing -padding	
7	DUT sends Unconfirmed frame	→	Retransmission fragment (lost tiles): Fport,=FPortUp W=0, FCN=computed , and bytes from first-window lost tiles or from first-window lost tiles and second-window lost tiles	
	TCL sends Unconfirmed frame	←	SCHC ACK-incorrect RCS -RuleID=FPortUp -W(2 bits)=1 -C (1 bit)=0 -Compressed bitmap: binary depends on the tiles missing -padding Note: This occurs if tiles of windows 1 (second windows) are lost and not retransmitted in the previous retransmitted fragment also, otherwise SCHC ACK with correct RCS is sent	
8	DUT sends Unconfirmed frame	→	If the SCHC ACK was not sent in the previous step, Retransmission fragment (lost tiles): Fport,=FPortUp W=1, FCN=computed, and bytes from second-window lost tiles	
	TCL sends Unconfirmed frame	←	SCHC ACK RuleID: FPortUp W= value of last window C=1 Padding	
9	DUT sends Unconfirmed frame	→		

3.8 Uplink Fragmentation-Retransmission (variable MTU)

Test Purpose:

Test the capability of the DUT to retransmit when there is error/missing when the MTU is changing.

Test Description

This test is the same as Uplink fragmentation (variable MTU) Section 3.6, with the difference that if some tiles are missing, the TCL will reply to the DUT by sending a bitmap.

Note: There are two possibilities for SCHC to send a bitmap – either at the end of each window or until the end of all windows. This is determined by the implementer. Here, the one at the end of the windows is used, as it will work with both.

3.8.1 Test Procedure Frame Sequence Chart

Step	Procedure	Frame Sequence		Test Purpose
		End Device - TCL	Frame	
1	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	MAC-CMD LinkADRReq DataRate = Max125kHzDR, refer [2] Payload = [0x]03XXXXXXXXXX	
2	DUT sends Unconfirmed frame	→	MAC-CMD LinkADRAns Payload = [0x]0307	
	TCL sends Unconfirmed frame	←	CP-CMD SCHCMsgSendReq FPort = 224 Payload = [0x]5070308A02	Test DUT's Fragmentation mechanism (MTU changing)
3	DUT sends Unconfirmed frame	→	CP-CMD SCHCMsgSendAns FPort = 224 Payload = [0x]50	
4	DUT sends Unconfirmed frame	→	First fragment: FPortUp in FPort, W=0, FCN=62, RuleIdx, bytes from (IPv6 header, UDP header, Data)*	
5	DUT sends Unconfirmed frame	→ R [until the last fragment]	Next (not last) fragment: FPortUp in Fport, W=computed, FCN=computed, and bytes from (IPv6 header, UDP header, Data)*	
6	DUT sends Unconfirmed frame	→	All-1 Fragment (last fragment) FPortUp in Fport, W=computed, FCN=B'111111, RCS=computed	The last tile can be held on regular SCHC fragment or in the All-1
	The TCL sends Unconfirmed frame	←	MAC-CMD LinkADRReq DataRate = Max125kHzDR-2, refer [2] Payload = [0x]03XXXXXXXXXX	
7	DUT sends Unconfirmed frame	→	MAC-CMD LinkADRAns Payload = [0x]0307	

	TCL sends Unconfirmed frame	←	SCHC ACK-incorrect RCS -RuleID=FPortUp -W(2 bits)=0 -C(1 bit)=0 -Compressed bitmap: binary depends on the missing tiles -Padding	
8	DUT sends Unconfirmed frame	→	Retransmission fragment (lost tiles): Fport=FPortUp W=0, FCN=computed, and bytes from first-window lost tiles or from first-window lost tiles and second-window lost tiles (may be divided into several fragments if the MTU decreased)	
	TCL sends Unconfirmed frame	←	SCHC ACK-incorrect RCS -RuleID=FPortUp -W(2 bits)=1 -C(1 bit)=0 -Compressed bitmap: binary depends on the missing tiles -Padding This occurs if tiles of windows 1 are lost also and not retransmitted in the previous retransmitted fragment, otherwise SCHC ACK with correct RCS is sent	
9	DUT sends Unconfirmed frame	→	If the SCHC ACK was not sent in the previous step, Retransmission fragment (lost tiles): Fport=FPortUp W=1, FCN=computed, and bytes from second-window lost tiles (may be divided into several fragments if the MTU decreased)	
	TCL sends Unconfirmed frame	←	SCHC ACK RuleID: FPortUp W= value of last window C=1 Padding	
10	DUT sends Unconfirmed frame	→		

3.9 Uplink Fragmentation- ACK Request retransmission

Test Purpose:

Test the behaviour of the DUT when the All-1 is lost (ACK request retransmission)

Test Description

This test is the same as Uplink fragmentation (fixed MTU) Section 3.5, with the difference that the TCL will not send a SCHC ACK. The DUT retransmission timer expires, and the DUT must retransmit the All-1 SCHC message (SCHC Ack request).

In this test, the TCL replies after receiving two SCHC ACK requests. The DUT then consider that the message is delivered.

3.9.1 Test Procedure Frame Sequence Chart

Step	Procedure	Frame Sequence		Test Purpose
		End Device - TCL	Frame	
1	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	MAC-CMD LinkADRReq DataRate = Max125kHzDR, refer [2] Payload = [0x]03XXXXXXXX	
2	DUT sends Unconfirmed frame	→	MAC-CMD LinkADRAns Payload = [0x]0307	
	TCL sends Unconfirmed frame	←	CP-CMD SCHCMsgSendReq FPort = 224 Payload = [0x]5070308A02	Test DUT's Fragmentation mechanism
3	DUT sends Unconfirmed frame	→	CP-CMD SCHCMsgSendAns FPort = 224 Payload = [0x]50	
4	DUT sends Unconfirmed frame	→	First fragment: FPortUp in FPort, W=0, FCN=62, RuleIdx, bytes from (IPv6 header, UDP header, Data)	
5	DUT sends Unconfirmed frame	→ R [until the last fragment]	Next (not last) fragment: FPortUp in Fport, W=computed, FCN=computed, and bytes from (IPv6 header, UDP header, Data)	
6	DUT sends Unconfirmed frame	→	All-1 Fragment FPortUp in Fport, W=computed, FCN=B'111111, RCS=computed	The last tile can be held on regular SCHC fragment or in the All-1

7	DUT sends Unconfirmed frame	→	After retransmission timer expires, the DUT sends a SCHC ACK request: FPortUp in Fport, W= of last window, FCN=0 (6 bits)	
8	DUT sends Unconfirmed frame	→	After retransmission timer expires, the DUT sends a SCHC ACK request: FPortUp in Fport, W= of last window, FCN=0 (6 bits)	
	TCL sends Unconfirmed frame	←	SCHC ACK RuleID: FPortUp W= value of last window C=1 Padding	
9	DUT sends Unconfirmed frame	→		

3.10 Uplink Fragmentation-Sender Abort (no SCHC ACK)

Test Purpose:

Test the behaviour of the DUT to avoid getting loop due to not receiving SCHC ACK.

Test Description

This test is the same as Uplink fragmentation (fixed MTU) Section 3.5, with the difference that the TCL will not send a SCHC ACK. The DUT retransmission timer expires and the DUT must retransmit the All-1 SCHC message (SCHC Ack request).

This must be repeated MAX_ACK_REQ times, after which the DUT must send a Sender Abort message

Note: The All-1 message is counted as a SCHC Ack REQ

3.10.1 Test Procedure Frame Sequence Chart

Step	Procedure	Frame Sequence		Test Purpose
		End Device - TCL	Frame	
1	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	MAC-CMD LinkADRReq DataRate = Max125kHzDR, refer [2] Payload = [0x]03XXXXXXXXXX	

2	DUT sends Unconfirmed frame	→	MAC-CMD LinkADRAAns Payload = [0x]0307	
	TCL sends Unconfirmed frame	←	CP-CMD SCHCMsgSendReq FPort = 224 Payload = [0x]5070308A02	Test DUT's Fragmentation mechanism
3	DUT sends Unconfirmed frame	→	CP-CMD SCHCMsgSendAns FPort = 224 Payload = [0x]50	
4	DUT sends Unconfirmed frame	→	First fragment: FPortUp in FPort, W=0, FCN=62, RuleIdx, bytes from (IPv6 header, UDP header, Data)*	
5	DUT sends Unconfirmed frame	→ R [until last fragment]	Next (not last) fragment: FPortUp in Fport, W=computed, FCN=computed, and bytes from (IPv6 header, UDP header, Data)	
6	DUT sends Unconfirmed frame	→	All-1 Fragment (last fragment) FPortUp in Fport, W=computed, FCN=b'111111, RCS=computed	The last tile can be held on regular SCHC fragment or in the All-1
7	DUT sends Unconfirmed frames	→ R [MAX_ACK_REQ times]	After retransmission timer expires, the DUT sends a SCHC ACK request. (Repeat MAX_ACK_REQ times) FPortUp in Fport, W= of last window, FCN=0 (6 bits)	
8	DUT sends Unconfirmed frames	→	After retransmission timer expires, and MAX_ACK_REQ number of All-1 is sent, the DUT sends the Sender Abort message FPortUp W=b'11 FCN= b'111111	
9	DUT sends Unconfirmed frame	→		

3.11 Uplink Fragmentation-Sender Abort (all tiles received but incorrect RCS)

Test Purpose:

Concluding a failed transmission (C=0 and no missing tiles)

Test Description

This test is the same as Uplink fragmentation (fixed MTU) Section 3.5, with the difference that the TCL will send a SCHC ACK with incorrect RCS and no missing tile.

The DUT must send a Sender Abort message.

3.11.1 Test Procedure Frame Sequence Chart

Step	Procedure	Frame Sequence		Test Purpose
		End Device - TCL	Frame	
1	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	MAC-CMD LinkADRReq DataRate = Max125kHzDR, refer [2] Payload = [0x]03XXXXXXXX	
2	DUT sends Unconfirmed frame	→	MAC-CMD LinkADRAns Payload = [0x]0307	
	TCL sends Unconfirmed frame	←	CP-CMD SCHCMsgSendReq FPort = 224 Payload = [0x]5070308A02	Test DUT's Fragmentation mechanism
3	DUT sends Unconfirmed frame	→	CP-CMD SCHCMsgSendAns FPort = 224 Payload = [0x]50	
3	DUT sends Unconfirmed frame	→	First fragment: FPortUp in FPort, W=0, FCN=62, RuleIDx, bytes from (IPv6 header, UDP header, Data)*	
4	DUT sends Unconfirmed frame	→ R [until last fragment]	Next (not last) fragment: FPortUp in Fport, W=computed, FCN=computed, and bytes from (IPv6 header, UDP header, Data)	
5	DUT sends Unconfirmed frame	→	All-1 Fragment (last fragment) FPortUp in Fport, W=computed, FCN=b'111111, RCS=computed	The last tile can be held on regular SCHC fragment or in the All-1
6	TCL sends Unconfirmed frame	←	SCHC ACK RuleID: FPortUp W= value of last window C=0 Bitmap=everything is received Padding	

7	DUT sends Unconfirmed frame	→	Sender Abort FPortUp W=b'11 FCN= b'111111	
8	DUT sends Unconfirmed frame	→		

3.12 Uplink Fragmentation-Receiver Abort

Test Purpose:

Stopping a transmission when the sender receives a receiver abort

Test Description

This test is the same as Uplink fragmentation (fixed MTU) Section 3.5, with the difference that the TCL sends a SCHC receiver abort as an imitation of inactivity timer expiration. Upon expiration of the Inactivity Timer, the TCL sends a SCHC Receiver-Abort.

The DUT must stop sending the fragments.

3.12.1 Test Procedure Frame Sequence Chart

Step	Procedure	Frame Sequence		Test Purpose
		End Device - TCL	Frame	
1	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	MAC-CMD LinkADRReq DataRate = Max125kHzDR, refer [2] Payload = [0x]03XXXXXXXX	
2	DUT sends Unconfirmed frame	→	MAC-CMD LinkADRAns Payload = [0x]0307	
	TCL sends Unconfirmed frame	←	CP-CMD SCHCMsgSendReq FPort = 224 Payload = [0x]5070308A02	Test DUT's Fragmentati on mechanism

3	DUT sends Unconfirmed frame	→	CP-CMD SCHCMsgSendAns FPort = 224 Payload = [0x]50	
4	DUT sends Unconfirmed frame	→	First fragment: FPortUp in FPort, W=0, FCN=62, RuleIdx, bytes from (IPv6 header, UDP header, Data)	
	TCL sends Unconfirmed frame	←	Receiver Abort FPortUp in FPort W=b'11 C=b'1 13 bits of 1 Note: This message should be sent as soon as the TCL receives the first fragment	
5	DUT must stop sending the fragments as soon as it receives the Receiver Abort message			
6	DUT sends Unconfirmed frame	→		

3.13 Uplink-Compression and fragmentation

Test Purpose:

Testing the capability of DUT to apply both fragmentation and compression in uplink.

Test Description

Input:

The TCL sends the DUT a CP-CMD SCHCMsgSendReq to ask the DUT to send an uplink message with a large data size = 650 bytes and 0x1070 as a UDP App Port.

Expected Output:

The DUT must reply with the SCHCMsgSendAns.

A compressed SCHC packet with FPort=RuleIDy (default is 100) must be sent back to the TCL via several fragments.

3.13.1 Test Procedure Frame Sequence Chart

Step	Procedure	Frame Sequence		Test Purpose
		End Device - TCL	Frame	
1	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	MAC-CMD LinkADRReq DataRate = Max125kHzDR, refer [2] Payload = [0x]03XXXXXXXXXX	
2	DUT sends Unconfirmed frame	→	MAC-CMD LinkADRAns Payload = [0x]0307	
	TCL sends Unconfirmed frame	←	CP-CMD SCHCMsgSendReq FPort = 224 Payload = [0x]5070108A02	Test DUT's Fragmentati on mechanism
3	DUT sends Unconfirmed frame	→	CP-CMD SCHCMsgSendAns FPort = 224 Payload = [0x]50	
4	DUT sends Unconfirmed frame	→	First fragment: FPortUp in FPort, W=0, FCN=62, RuleIDy, Compression residue, bytes from Data	
5	DUT sends Unconfirmed frame	→ R [until last fragment]	Next (not last) fragment: FPortUp in Fport, W=computed, FCN=computed, and bytes from (Data, padding)* This fragment type is repeated until the last fragment	
6	DUT sends Unconfirmed frame	→	All-1 Fragment FPortUp in Fport, W=computed, FCN=b'111111, RCS=computed	The last tile can be held on regular SCHC fragment or in the All-1
	TCL sends Unconfirmed frame	←	SCHC ACK RuleID: FPortUp W= value of last window C=1 Padding	
7	DUT sends Unconfirmed frame	→		

3.14 Downlink fragmentation- reassembly

Test Purpose:

Testing the capability of DUT to reassemble downlink fragments

Test Description

Input: The TCL initiates a downlink fragmentation session on FPort FPortDwn

The TCL sends 3 fragments to the DUT:

1. Fragment 1: Fport=FportDwn, W=0 (1 bit), FCN=0 (1 bit), RuleID=RuleIDx (1 byte), 28 bytes +6 bits of (IPv6 and UDP header/data)
2. Fragment 2: Fport=FportDwn, W=1 (1 bit), FCN=0 (1 bit), 29 bytes +6 bits of (IPv6 and UDP header/data)
3. Fragment 3: Fport=FportDwn, W=0 (1 bit), FCN=1 (1 bit), RCS (4 bytes), 25 bytes +6 bits of (IPv6 and UDP header/data) including padding

Expected Output: Upon receiving the fragments, the DUT must acknowledge each packet (ACK-Always), reassemble the SCHC packet and resend it on FPort=224 using the SCHCACKInd message.

3.14.1 Test Procedure Frame Sequence Chart

Step	Procedure	Frame Sequence		Test Purpose
		End Device - TCL	Frame	
1	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	MAC-CMD LinkADRReq DataRate = Max125kHzDR, refer [2] Payload = [0x]03XXXXXXXXXX	
2	DUT sends Unconfirmed frame	→	MAC-CMD LinkADRAns Payload = [0x]0307	
	TCL sends Unconfirmed frame	←	First fragment: FPort FPortDwn W=0 (1 bit) FCN=0 (1 bit) RuleID=RuleIDx (1 byte) 28 bytes +6 bits of data	
3	DUT sends Unconfirmed frame	→	SCHC ACK RuleID FPortDwn W=0 (1 bit) C=0 (1 bit) b=1 (1 bit) Padding (5 bits)	

	TCL sends Unconfirmed frame	←	Second fragment: FPort FPortDwn W=1 (1 bit) FCN=0 (1 bit) 29 bytes +6 bits of data	
4	DUT sends Unconfirmed frame	→	SCHC ACK RuleID FPortDwn W=1 (1 bit) C=0 (1 bit) b=1 (1 bit) Padding (5 bits)	
	TCL sends Unconfirmed frame	←	Last fragment: FPort FPortDwn W=0 (1 bit) FCN=1 (1 bit) RCS (4 bytes)=computed 25 bytes +6 bits of data	
5	DUT sends Unconfirmed frame	→	SCHC ACK RuleID FPortDwn W=0 (1 bit) C=1 (1 bit) Padding (6 bits)	
6	DUT sends Unconfirmed frame	→	CP-CMD SCHCACKInd FPort = 224 Payload=[0x]51 48 bytes of IPv6/UDP header+36 bytes of payload	
7	DUT sends Unconfirmed frame	→		

3.15 Downlink fragmentation-Receiver abort

Test Purpose:

Test special behaviour of DUT when it stops receiving data for more than inactivity timer.

Test Description

Input: The TCL initiates downlink fragmentation on FPort FPortDwn. After sending the first fragment (windows), the TCL stops sending any more fragments.

Expected Output:

After receiving the first fragment, when the DUT does not receive any more fragments for longer than SCHC inactivity timer or after the DUT sends SCHC ACK for MAX_ACK_REQ times, the DUT must stop the reassembly process and send Receiver Abort message.

3.15.1 Test Procedure Frame Sequence Chart

Step	Procedure	Frame Sequence		Test Purpose
		End Device - TCL	Frame	
1	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	MAC-CMD LinkADRRReq DataRate = Max125kHzDR, refer [2] Payload = [0x]03XXXXXXXXXX	
2	DUT sends Unconfirmed frame	→	MAC-CMD LinkADRAns Payload = [0x]0307	
	TCL sends Unconfirmed frame	←	First fragment: FPort FPortDwn W=0 (1 bit) FCN=0 (1 bit) RuleID=RuleIDx (1 byte) 28 bytes +6 bits of data	
3	DUT sends Unconfirmed frame	→	SCHC ACK RuleID FPortDwn W=0 (1 bit) C=0 (1 bit) b=1 (1 bit) Padding (5 bits) Note: this step can be repeated for MAX_ACK_REQ time	
4	DUT sends Unconfirmed frame	→	After inactivity_timer expires or DUT sends MAX_ACK_REQ time of SCHC ACK, the DUT sends Receiver Abort Fport = FPortDwn W=`1` (1 bit) C=`1` (1 bit) `14 bits of `1s	
5	DUT sends Unconfirmed frame	→		

3.16 Downlink fragmentation-Lost Ack

Test Purpose:

Test the retransmission of ACK after being lost.

Test Description

Input: The TCL initiates downlink fragmentation on FPort FPortDwn.

The TCL will send 3 fragments to the DUT:

1. Fragment 1: Fport=FportDwn, W=0 (1 bit), FCN=0 (1 bit), RuleID=RuleIDx (1 byte), 28 bytes +6 bits of (IPv6 and UDP header/data)
2. Fragment 2: Fport=FportDwn, W=1 (1 bit), FCN=0 (1 bit), 29 bytes +6 bits of (IPv6 and UDP header/data)
3. Fragment 3: Fport=FportDwn, W=0 (1 bit), FCN=1 (1 bit), RCS (4 bytes), 25 bytes +6 bits of (IPv6 and UDP header/data) including padding

Data is IPv6/UDP headers (48 bytes) and 36 bytes payload and 2 bits of padding.

After the second fragment, the TCL behaves as if an ACK is lost and sends an ACK REQ.

Expected Output:

The DUT must send an ACK for each fragment.

The DUT must retransmit the ACK for the ACK REQ and continue reassembling the received packets.

Finally, the DUT must send a CP-CMD SCHCACKInd with Payload = 48 bytes of IPv6/UDP header+36 bytes of payload

3.16.1 Test Procedure Frame Sequence Chart

Step	Procedure	Frame Sequence		Test Purpose
		End Device - TCL	Frame	
1	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	MAC-CMD LinkADRReq DataRate = Max125kHzDR, refer [2] Payload = [0x]03XXXXXXXXXX	

2	DUT sends Unconfirmed frame	→	MAC-CMD LinkADRAAns Payload = [0x]0307	
	TCL sends Unconfirmed frame	←	First fragment: FPort FPortDwn W=0 (1 bit) FCN=0 (1 bit) RuleID=RuleIDx (1 byte) 28 bytes +6 bits of data	
3	DUT sends Unconfirmed frame	→	SCHC ACK RuleID FPortDwn W=0 (1 bit) C=0 (1 bit) b=1 (1 bit) Padding (5 bits)	
	TCL sends Unconfirmed frame	←	Second fragment: FPort FPortDwn W=1 (1 bit) FCN=0 (1 bit) 29 bytes +6 bits of data	
4	DUT sends Unconfirmed frame	→	SCHC ACK RuleID FPortDwn W=1 (1 bit) C=0 (1 bit) b=1 (1 bit) Padding (5 bits)	
	TCL sends Unconfirmed frame	←	ACK REQ (after retransmission timer) FPortDwn W=1 FCN=0 Padding	
5	DUT sends Unconfirmed frame	→	SCHCACK RuleID FPortDwn W=1 (1 bit) C=0 (1 bit) b=1 (1 bit) Padding (5 bits)	
	TCL sends Unconfirmed frame	←	Last fragment: FPort FPortDwn W=0 (1 bit) FCN=1 (1 bit) RCS (4 bytes) = computed 25 bytes +6 bits of data	
6	DUT sends Unconfirmed frame	→	SCHC ACK RuleID FPortDwn W=0 (1 bit) C=1 (1 bit) Padding (6 bits)	
7	DUT sends Unconfirmed frame	→	CP-CMD SCHCACKInd FPort = 224 Payload =[0x]51 48 bytes of IPv6/UDP header+36 bytes of payload	
8	DUT sends Unconfirmed frame	→		

3.17 Downlink fragmentation (reassembly) and decompression

Test Purpose:

Testing the capability of DUT to reassemble downlink fragments and decompress SCHC packet.

Test Description

Input: The TCL will initiate downlink fragmentation session on FPort FPortDwn of a compressed SCHC packet using RuleIDz.

The TCL sends 2 fragments to the DUT:

1. Fragment 1: Fport=FportDwn, W=0 (1 bit), FCN=0 (1 bit), RuleID=RuleIDz (1 byte), Compression residue (b'11100000000), 27 bytes + 3 bits of data
2. Fragment 2: Fport=FportDwn, W=1 (1 bit), FCN=1 (1 bit), RCS (4 bytes), 24 bytes + 6 bits of data (and padding)

Expected Output: Upon receiving the fragments, the DUT must acknowledge each packet (ACK-Always), reassemble the SCHC packet, decompress it and resend it on FPort=224 using the SCHCACKInd message,

Decompressed SCHC packet:

40 bytes IPv6, 8 bytes UDP, 52 bytes of Data

3.17.1 Test Procedure Frame Sequence Chart

Step	Procedure	Frame Sequence		Test Purpose
		End Device - TCL	Frame	
1	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	MAC-CMD LinkADRRReq DataRate = DR4, refer [2] Payload = [0x]03XXXXXXXXXX	

2	DUT sends Unconfirmed frame	→	MAC-CMD LinkADRs Payload = [0x]0307	
	TCL sends Unconfirmed frame	←	First fragment: Fport=FportDwn W=0 (1 bit) FCN=0 (1 bit), RuleID=RuleIDz (1 byte), Compression residue (b'11100000000) 27 bytes +3 bits of data	
3	DUT sends Unconfirmed frame	→	SCHC ACK RuleID FPortDwn W=0 (1 bit) C=0 (1 bit) b=1 (1 bit) Padding (5 bits)	
	TCL sends Unconfirmed frame	←	Last fragment: FPort FPortDwn W=1 (1 bit) FCN=1 (1 bit) RCS (4 bytes)=computed 25 bytes +6 bits of data	
4	DUT sends Unconfirmed frame	→	SCHC ACK RuleID FPortDwn W=1 (1 bit) C=1 (1 bit) Padding (6 bits)	
5	DUT sends Unconfirmed frame	→	CP-CMD SCHCACKInd FPort = 224 Payload = [0x]51 and 48 bytes of IPv6/UDP header+52 bytes of payload	
6	DUT sends Unconfirmed frame	→		